BSH/W-23/CS/IT/AI/DS1T002/02

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU& RTMNU

Education to Eternity

B.Tech. Winter - 2023

(ESE/ Make-up / Supplementary) Examination

Branch: CSE/AI/CSE-DS **Sem:-I**

Subject: Advanced Physics **Subject Code: CS1T002/ AI1T002/ DS1T002** IT1T002 **Marks: 100**

Date of Exam: 29/12/2023 **Time: 3 Hrs**

Instructions:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.
4. Use of Programmable Calculator is prohibited.
5. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.

COs: Students should be able to:

CO1: Describe the concept of LASER, optical fiber, types of semiconductors, PN junction diode characteristics, wave optics, electron Ballistics, quantum mechanics, various crystal structure parameters & X-rays.

CO2: Elaborate the types of LASER, optical fiber, Semiconductors, crystal structure, formation of Newton's ring, fringes in wedge shape thin film, effect of electric and magnetic field on motion of charge particle and significance of quantum mechanics.

CO3: Apply the concept of three and four level in LASER production, TIR in Optical fiber, classify the type of material based on current conduction, Bragg's law and X-ray diffraction, Interference for advanced application, illustrate the wave particle dualism of matter waves, motion and charged particle in E and B.

CO4: Analyze the different types of LASER and optical fiber, correlate the motion of charged particles in uniform electric and magnetic fields for Bainbridge Mass spectrograph, the formation of fringes in thin film, behaviour of wave function and the types of crystal.

CO5: Justify physical significance of wave function, HUP, Schrodinger's wave equations, application of Hall effect, LASER & Optical Fibre, Wave Optics, Electron Ballistics and interpret he various crystal structure.

CO6: Design devices by using the concept of Laser, optical fibre, Electron ballistics, Semiconductor, crystals structure, wave optics and quantum mechanics.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Explain the principal on which optical signal propagate through Optical fibre?	2/CO2	2
Q.2	Name the doping material for preparation of N type Semiconductor.	1/CO1	2
Q.3	Why the central spot is dark in Newton's Ring obtained due reflected light.	1/CO1	2
Q.4	Clarify the role of crossed field configuration as velocity filter?	1/CO1	2
Q.5	What are specifications of Simple cubic cell?	2/CO2	2
Q.6	What is the matter wave?	1/CO1	2
Q.7	At what thickness of thin film, reflected rays shows constructive interference pattern?	1/CO1	2
Q.8	What is co-ordination number for body centred cubic cell?	1/CO1	2
Q.9	What is mean of pumping in LASER?	1/CO1	2
Q.10	Justify the state of electron in uniform Magnetic field B when it is at rest initially.	2/CO2	2
Q.11	Explain, light is dual in nature.	1/CO1	2
Q.12	What is nos. of atoms per unit cell(Z) for FCC type of Cell	1/CO1	2

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SECTION B

All Questions are compulsory

Q.13	Explain the following transition process with neat sketches and diagram 1)Spontaneous emission 2) Stimulated emission	2/CO2	4
Q.14	With neat sketches and band diagram, compare P and N-type semiconductor at room temperature.	3/CO3	4
Q.15	Why the Newton's Rings are circular? Comment. Calculate the radius of curvature of plano-convex lens in Newton's ring experiment if the radius of 10th dark ring is 5.2×10^{-4} m and monochromatic light of wavelength of 500×10^{-9} m is used.	3/CO3	4
Q.16	Explain the role of velocity Selector in Bainbridge mass spectrograph. Determine the magnitude of magnetic field B of velocity selector if the ions passing through it is un-deflected with velocity 5×10^{3} m/s and electric filed applied across the plates is 500V/m..	3/CO3	4
Q.17	What is de'Broglie hypothesis? Develop an equation for de'Broglie wavelength for an electron accelerated through potential of V volts is $\frac{12.26}{\sqrt{V}}$ Å	2/CO2	4
Q.18	What is Miller Indices? Draw following crystal planes in cubical cell having Miller Indices. i) (100), ii)(101) iii)(121)	2/CO2	4

SECTION C

Solve Any FOUR Questions.

Q.19	What is Fermi energy? Show that Fermi energy is lies in the middle of Band gap of an intrinsic semiconductor with expression and neat sketches.	3/CO3	7
Q.20	What is fringe width? Obtain an expression for fringe width in a wedge shape thin film experiment. Comment, why the fringes are straight and parallel?	3/CO3	7
Q.21	Comment, how electron follows helical path in uniform Magnetic field? Derive an expression for radius of Circular path in helix, Pitch of the helix and Time period of revolution of circular path in helix.	4/CO4	7
Q.22	What is Heisenberg Uncertainty Principal? Show that electron cannot exist inside the nucleus.	3/CO3	7
Q.23	Define the term Inter-planner spacing d_{hkl}. Derive an expression for it in Cubic unit cell. Calculate it for BCC cell having atomic radius 1.565 Å for plane of Miller indices (121).	4/CO4	7
Q.24	Explain the construction and working of the Ruby LASER with its energy level diagram and neat sketches.	3/CO3	7

SECTION D

Solve any TWO Questions

Q.25	a)Define the term i)Acceptance angle ii)Acceptance cone iii)Numerical Aperture iv) Fractional Refractive index change (Δ). Show that Numerical aperture N.A $= n_1\sqrt{2.\Delta}$	3/CO3	12
	b)A step-index fibre has a core and cladding R.I. of 1.5 and 1.4 respectively, calculate N.A. and Acceptance angle.		
Q.26	a)What is Hall Effect? Derive an expression for Hall coefficient, Hall voltage, Hall angle and Mobility with necessary diagram for P-type semiconductor. Write one application of Hall effect.		
	b)A P-type germanium sample has a Acceptor density of 10^{21} m^{-3}. It is arranged in Hall experiment having magnetic field of 0.15 Tesla. If current of 2.5×10^{-2} Amp is passed through a sample of thickness 3 mm, find the Hall coefficient and Hall voltage.	4/CO4	12
Q.27	a)What is wave function Ψ? Derive an expression for Schrodinger's time independent wave equation of motion.		
	b)Calculate the de'Broglie wavelength of wave associated with particle of mass 1 gm and electron moving with velocity of 3×10^{6} m/s. Give your conclusion for both particles.	4/CO4	12

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